Exploratory Data Analysis

Comparison of the FB15k-237 and YAGO3-10 knowledge graphs

1. Objective and scope

The analysis examined data quality and graph structure before investigating whether embeddings capture knowledge graph semantics. The local training files **fb15k-237_train.tsv** and **yago3-10_train.csv** were analysed. Each row represents a directed triple (head entity, relation, tail entity).

The checks covered missing values, duplicates, relation frequencies, entity degrees, connectivity and relational profiles. Statistics were computed for the complete files after validation and deduplication. Entity degree is the sum of incoming and outgoing triple counts; a self-loop contributes two to this sum.

2. Basic statistics

Metric	FB15k-237	YAGO3-10
Number of triples	272,115	1,079,040
Number of entities	14,505	123,143
Number of relations	237	37
Missing values / redundant duplicate triples	0 / 0	0 / 0
Self-loops	1,625	0
Mean entity degree	37.52	17.52
Median entity degree	22	10
Entities with degree ≤ 5	11.1%	25.6%
Share of triples in the five most frequent relations	22.6%	82.9%

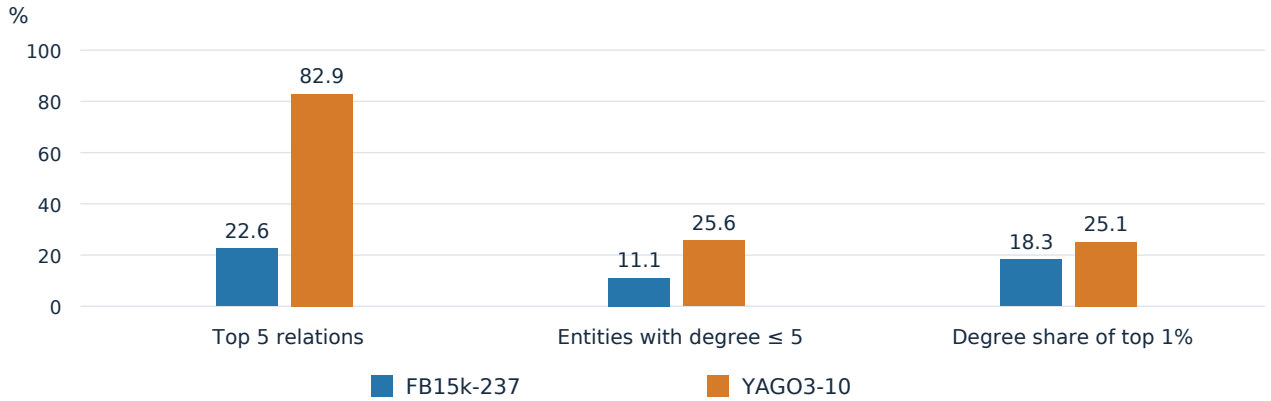

Figure 1. Relation and degree concentration. The last pair of bars shows the share of the total degree contributed by the 1% of entities with the highest degrees.

3. Relation and entity degree distributions

YAGO3-10 contains approximately four times as many triples as FB15k-237, but far fewer relations. Its relation distribution is highly imbalanced: the five most frequent relations account for 82.9% of triples. The most frequent relation, **isAffiliatedTo**, accounts for 34.6%. In FB15k-237, the most frequent relation accounts for 5.9%, and the top five together account for 22.6%. Model performance averaged over triples may therefore depend more strongly on a few dominant relations in YAGO.

In both datasets, the mean degree exceeds the median, indicating an uneven distribution of connections. The 1% of entities with the highest degrees contribute 18.3% of the total degree in FB15k-237 and 25.1% in YAGO3-10. At the same time, entities with degree at most five account for 11.1% and 25.6% of vertices, respectively. Highly connected hubs thus coexist with many entities that have limited structural context.

4. Connectivity and relation cardinality

When edge direction is ignored, both graphs are almost entirely connected: their largest weakly connected components contain 99.94% of FB15k-237 entities and 99.97% of YAGO3-10 entities. Accounting for direction reveals a substantial difference: the largest strongly connected components contain 80.1% and 5.7% of entities, respectively. Mutual reachability through directed paths is therefore much more limited in YAGO.

Triples belonging to many-to-many relations dominate: they account for 75.2% in FB15k-237 and 86.2% in YAGO3-10. A relation was assigned to this category when both the average number of distinct tails per head and the average number of distinct heads per tail were at least 1.5. This is an empirical property of the training data, rather than an ontological axiom.

5. Entity relational profiles

An entity profile was defined as a set of (direction, relation) pairs, excluding frequencies and neighbour identities. FB15k-237 yielded 10,642 distinct profiles, with 31.7% of entities sharing their profile with another entity. YAGO3-10 yielded only 4,646 profiles across 123,143 entities; 97.9% of entities shared their profile. YAGO therefore exhibits much greater repetition in this simplified description. This does not establish that these entities share a semantic class or have identical embeddings.

6. Conclusions and limitations

The datasets contain no missing values or identical triples requiring removal. The main differences concern relation concentration, the proportion of rare entities and profile repetition. Subsequent experiments should compare models with a simple structural representation and report results separately for frequent and rare entities and relations.

The analysis covers only the local training data. No recognized type predicates were found; evaluating classification or clustering against semantic classes requires additional labels. Without embeddings and validation and test splits, model quality and cross-split leakage cannot be assessed. This EDA provides preparation for that evaluation.

Source of results: data/fb15k_eda.ipynb and data/yago_eda.ipynb; local training files listed on page 1.